

Boolean Algebra Assignment

Question 1

Simplify this Boolean expression:

$$F = A'B'C' + A'B'C + A'BC + AB'C + ABC'$$

First I looked for terms that are almost the same.

$$F = (A'B'C' + A'B'C) + A'BC + AB'C + ABC'$$

The first two terms can be simplified because:

$$A'B'C' + A'B'C = A'B'$$

So now it becomes:

$$F = A'B' + A'BC + AB'C + ABC'$$

Then I factored out A' from the first two terms:

$$= A'(B' + BC) + AB'C + ABC'$$

Using Boolean algebra rule:

$$B' + BC = B' + C$$

So:

$$= A'(B' + C) + AB'C + ABC'$$

Expand again:

$$= A'B' + A'C + AB'C + ABC'$$

Then I grouped:

$$A'C + AB'C$$

Factor out C:

$$= C(A' + AB')$$

Using another rule:

$$A' + AB' = A' + B'$$

So:

$$= C(A' + B')$$

Expand:

$$= A'C + B'C$$

Now put everything back together:

$$F = A'B' + A'C + B'C + ABC'$$

That is the simplified expression.

Final Answer:

$$F = A'B' + A'C + B'C + ABC'$$

Motion Sensing Light System

Inputs:

m = motion sensed

t = test mode

Output:

i = lamp illuminate

The lamp should turn on when:

- motion is sensed and test mode is OFF
- test mode is ON and no motion
- test mode is ON and motion sensed

Original Boolean expression:

$$i = mt' + m't + mt$$

Now simplify it.

First group similar terms:

$$i = m(t' + t) + m't$$

Since:

$$t' + t = 1$$

Then:

$$i = m(1) + m't$$

$$i = m + m't$$

Using Boolean rule:

$$X + X'Y = X + Y$$

So:

$$i = m + t$$

Final simplified answer:

$$i = m + t$$

This means the lamp turns on if there is motion OR if test mode is on.

Flowchart Idea

Start



Read Boolean expression



Group similar terms



Apply Boolean algebra rules



Simplify expression



Check if more simplification possible

↓
Write final answer

↓
End

Challenges

The one thing I struggled with was the fact that some of the terms resembled one another but were not necessarily the same and couldn't be simplified in the same manner as the Boolean algebra rule. One other thing that was a bit confusing was setting up the expressions correctly and not dropping a variable. The part of the flowchart that is the chart itself was easier but still took some time to get in the correct order. From a few practices it was easy to start to simplify the expressions step by step.